Mixed-Factorial ANOVA Demo

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Mixed Factorial ANOVA | Research Context Prompt

The same stress researcher from our previous demonstration conducted another study to determine the impact of psychosocial stress on participants processing speed ability using a mixed-factorial research design.

The researcher randomly assigned 30 participants to one of two stress conditions: (1) Low Stress and (2) High Stress. Participants were assessed at three separate times across their academic career (Sophomore, Junior, and Senior) measuring a new processing speed performance measure. The dependent measure in this task was processing speed ability measured in milliseconds (500 ms = .5 seconds).

Again, this researcher believed that regulating feelings of stress would impair concurrent performance of processing speed and that these effects would materialize across time.

3 x 2 Mixed Factorial ANOVA | Data

```
library(psych)
library(tidyverse)
library(jmv)
library(ggpubr)
library(ggrain)
library(apaTables)
library(ez)
library(rstatix)

dat_MF <- read.csv("Mixed_Factorial.csv")

# SUPER IMPORTANT FOR WITHIN SUBJECT DESIGNS
# Set Subject or ID variable to a factor
dat_MF$Subject <- factor(dat_MF$Subject)

# Set IVs to factors
dat_MF$Condition <- factor(dat_MF$Condition, labels = c("Low-Stress","High-Stress"))
dat_MF$Rank <- factor(dat_MF$Rank, labels = c("Senior", "Junior", "Sophomore"))

# Assess number of participants and variables in data set
dim(dat_MF)
```

```
## [1] 90  4
```

```
head(dat_MF)
```

```
##   Subject   Condition      Rank Time
## 1       1  Low-Stress    Senior  512
## 2       1  Low-Stress    Junior  541
## 3       1  Low-Stress Sophomore  603
## 4       2 High-Stress    Senior  510
## 5       2 High-Stress    Junior  554
## 6       2 High-Stress Sophomore  596
```

```
# To use anovaRM() we must reshape our data set from `long` to `wide`
dat_MFW <- dat_MF %>%
  pivot_wider(names_from = Rank,
              values_from = Time)

# Let's Calculate an Average Time variable for assumption checking
dat_MFW$averageTime <- (dat_MFW$Sophomore + dat_MFW$Junior + dat_MFW$Senior) / 3
```

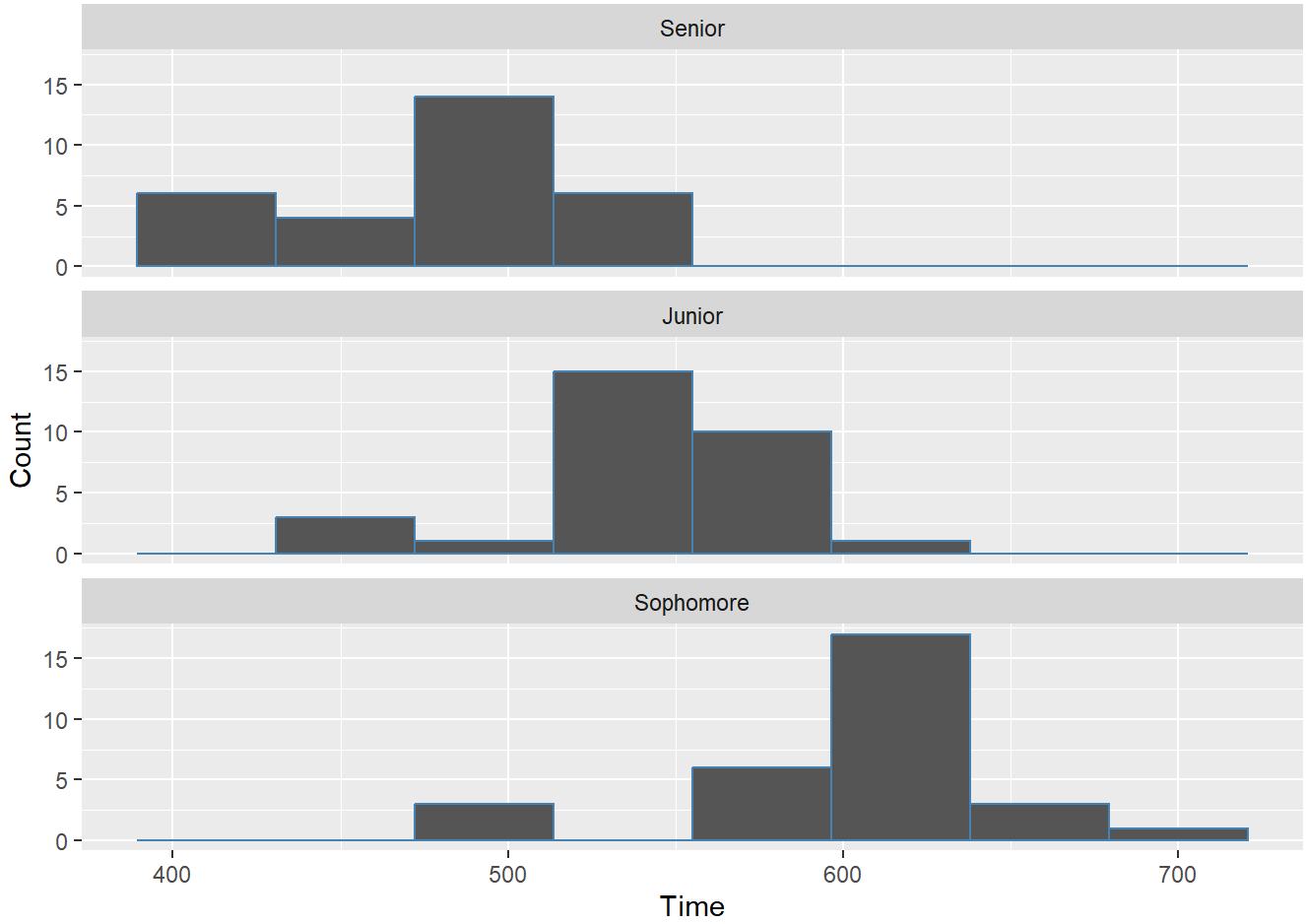
Mixed Factorial ANOVA | Descriptive Stats

```
# Descriptive Stats
describeBy(dat_MF$Time,
           list(dat_MF$Condition, dat_MF$Rank),
           mat=TRUE)
```

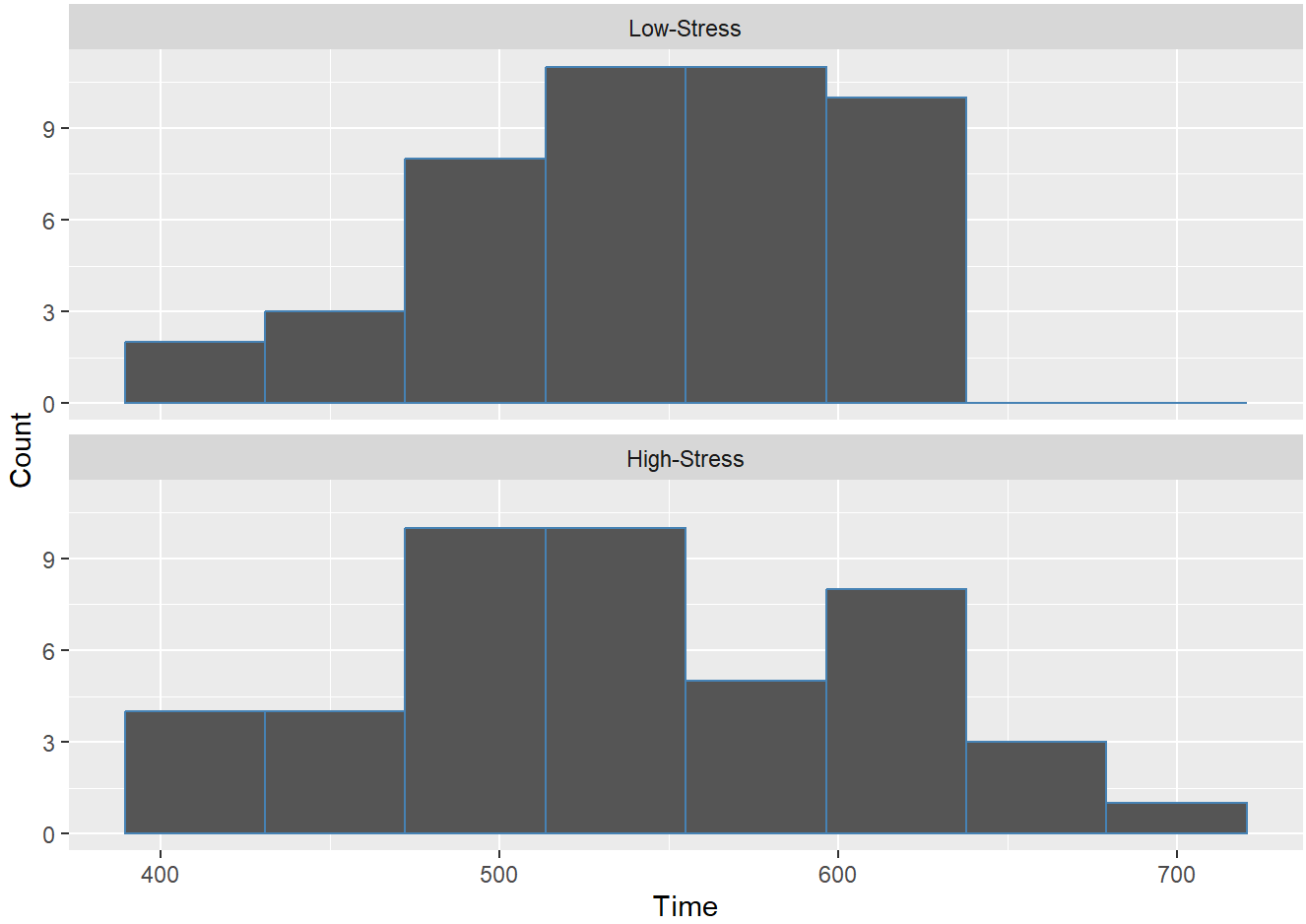
```
##     item      group1    group2 vars  n     mean       sd median  trimmed
## X11    1  Low-Stress    Senior    1 15 496.7333 40.83672    512 500.5385
## X12    2 High-Stress    Senior    1 15 465.6000 36.12834    480 466.3077
## X13    3  Low-Stress    Junior    1 15 543.6000 46.65650    559 547.8462
## X14    4 High-Stress    Junior    1 15 535.8667 26.90955    537 539.1538
## X15    5  Low-Stress Sophomore    1 15 587.3333 33.77164    598 591.4615
## X16    6 High-Stress Sophomore    1 15 618.4667 42.05416    616 620.5385
##         mad min max range       skew  kurtosis        se
## X11  8.8956 412 532   120 -1.2901196 -0.135849 10.543997
## X12 44.4780 410 512   102 -0.1840255 -1.659049  9.328298
## X13 28.1694 432 600   168 -1.1918834  0.467006 12.046655
## X14 23.7216 460 569   109 -1.2570249  1.603398  6.948016
## X15  8.8956 503 618   115 -1.7747683  1.588227  8.719800
## X16 26.6868 510 700   190 -0.5465190  1.044874 10.858337
```

Mixed Factorial ANOVA | Histogram

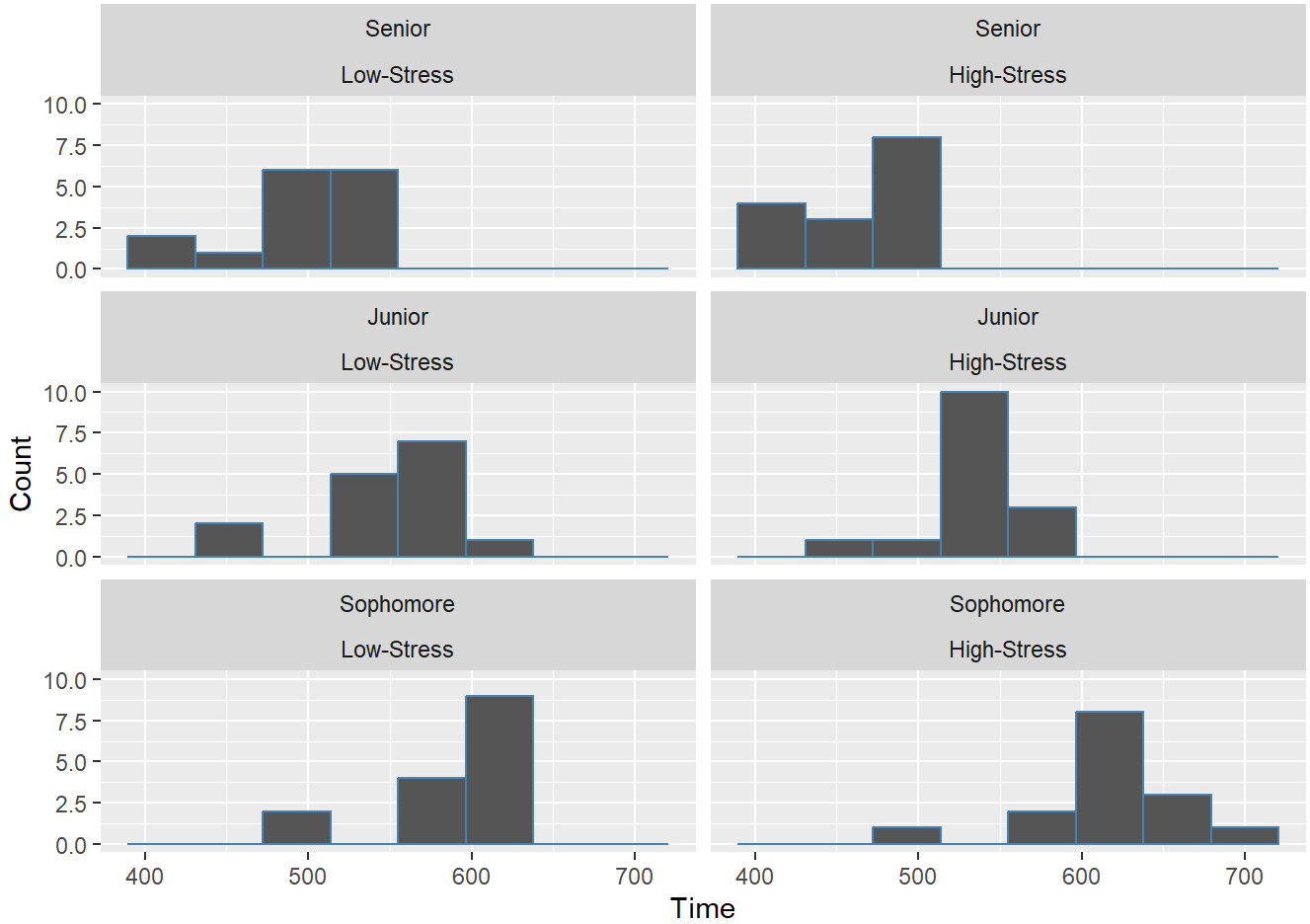
```
ggplot(data = dat_MF,
       mapping = aes(x = Time)) +
  geom_histogram(bins = 8, color = "steelblue") +
  labs(y = "Count", x = "Time")+
  facet_wrap(~dat_MF$Rank, ncol =1)
```




```
ggplot(data = dat_MF,
       mapping = aes(x = Time)) +
  geom_histogram(bins = 8, color = "steelblue") +
  labs(y = "Count", x = "Time")+
  facet_wrap(~dat_MF$Condition, ncol=1)
```



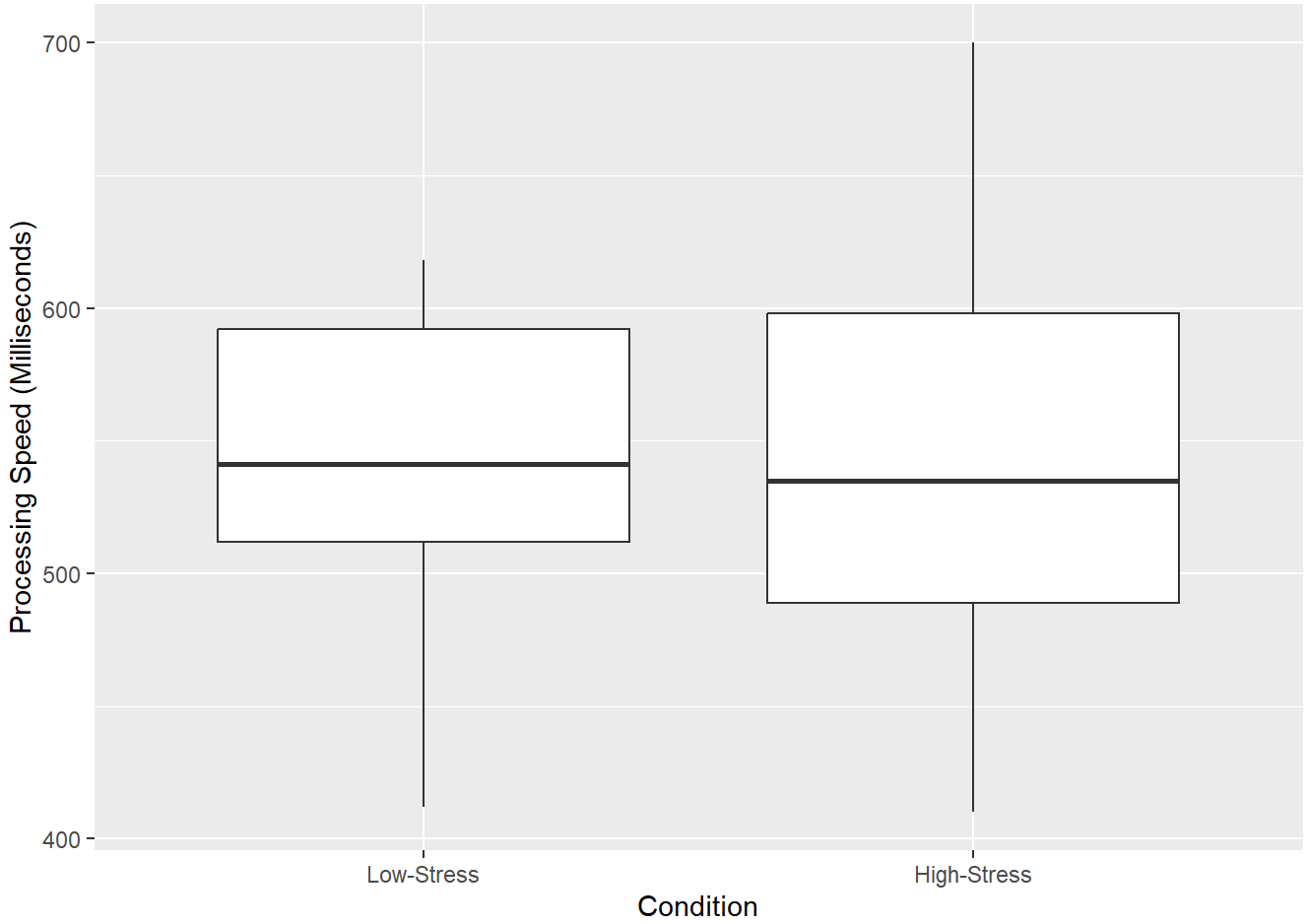

```
ggplot(data = dat_MF,
       mapping = aes(x = Time)) +
  geom_histogram(bins = 8, color = "steelblue") +
  labs(y = "Count", x = "Time")+
  facet_wrap(~dat_MF$Rank + dat_MF$Condition, ncol=2)
```



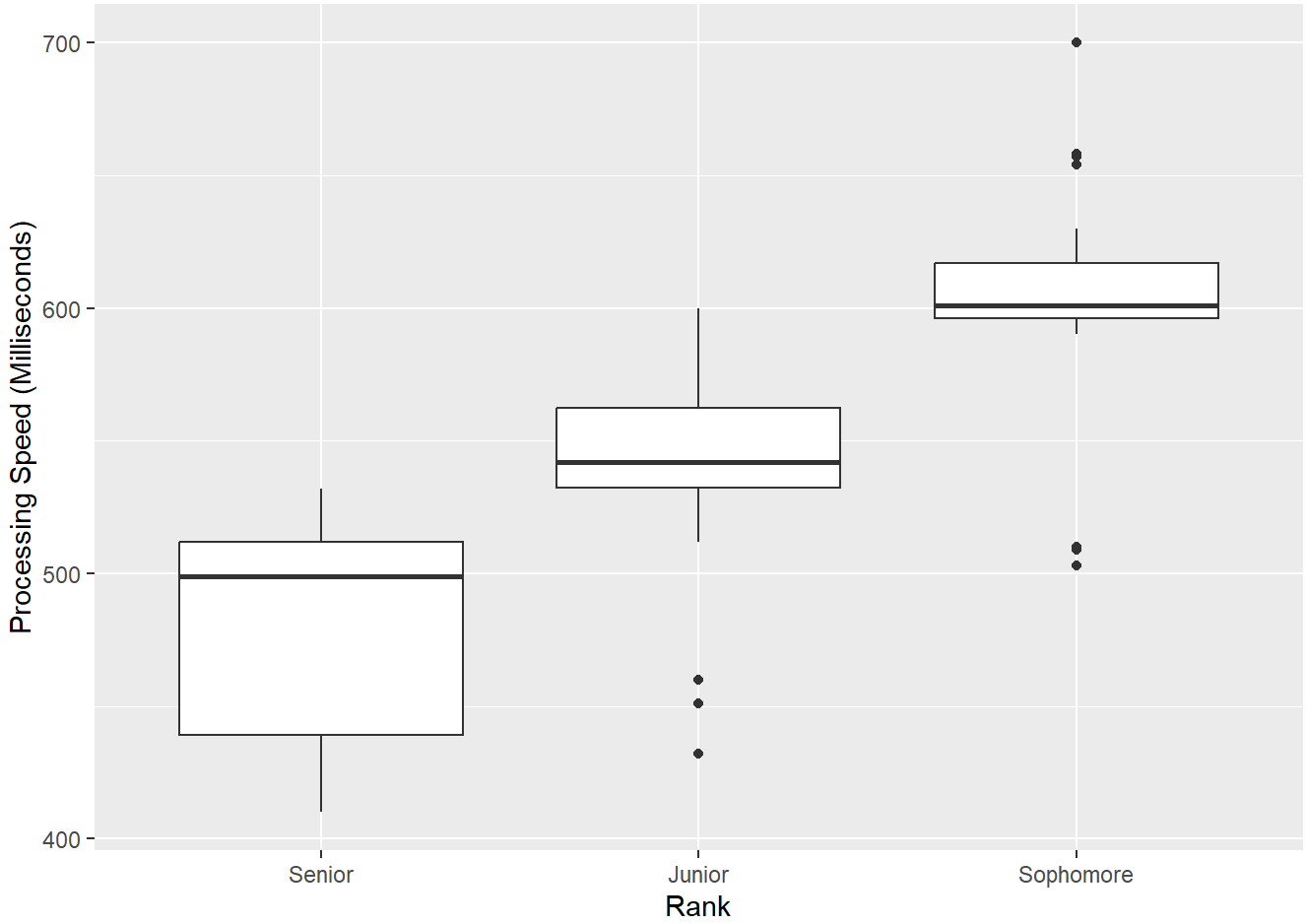
####

Mixed Factorial ANOVA | Boxplot and Raincloud Plot Visualiztaions

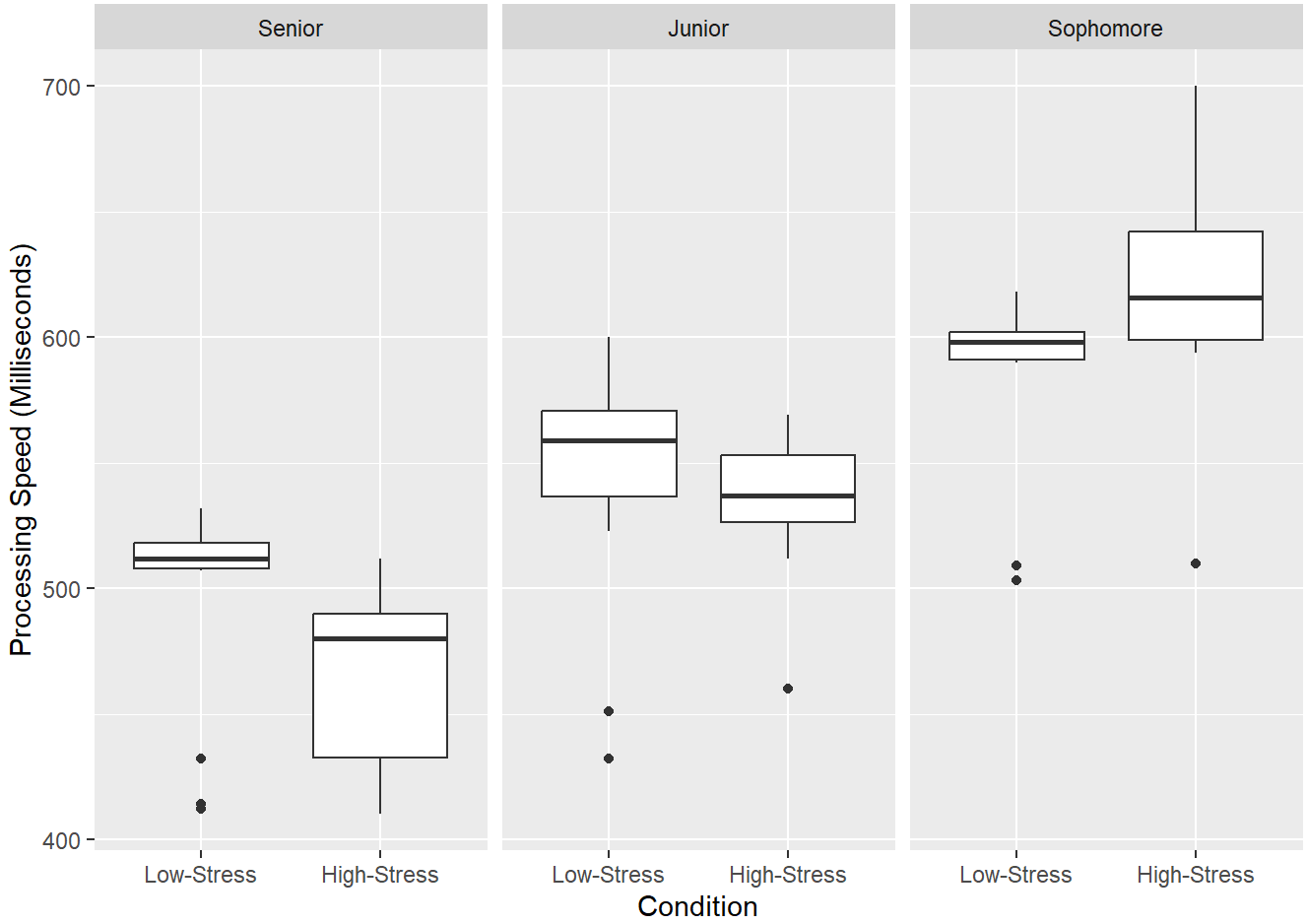
```
ggplot(data = dat_MF,
       mapping = aes(y = Time, x = Condition)) +
  geom_boxplot() +
    labs(y = "Processing Speed (Milliseconds)")
```



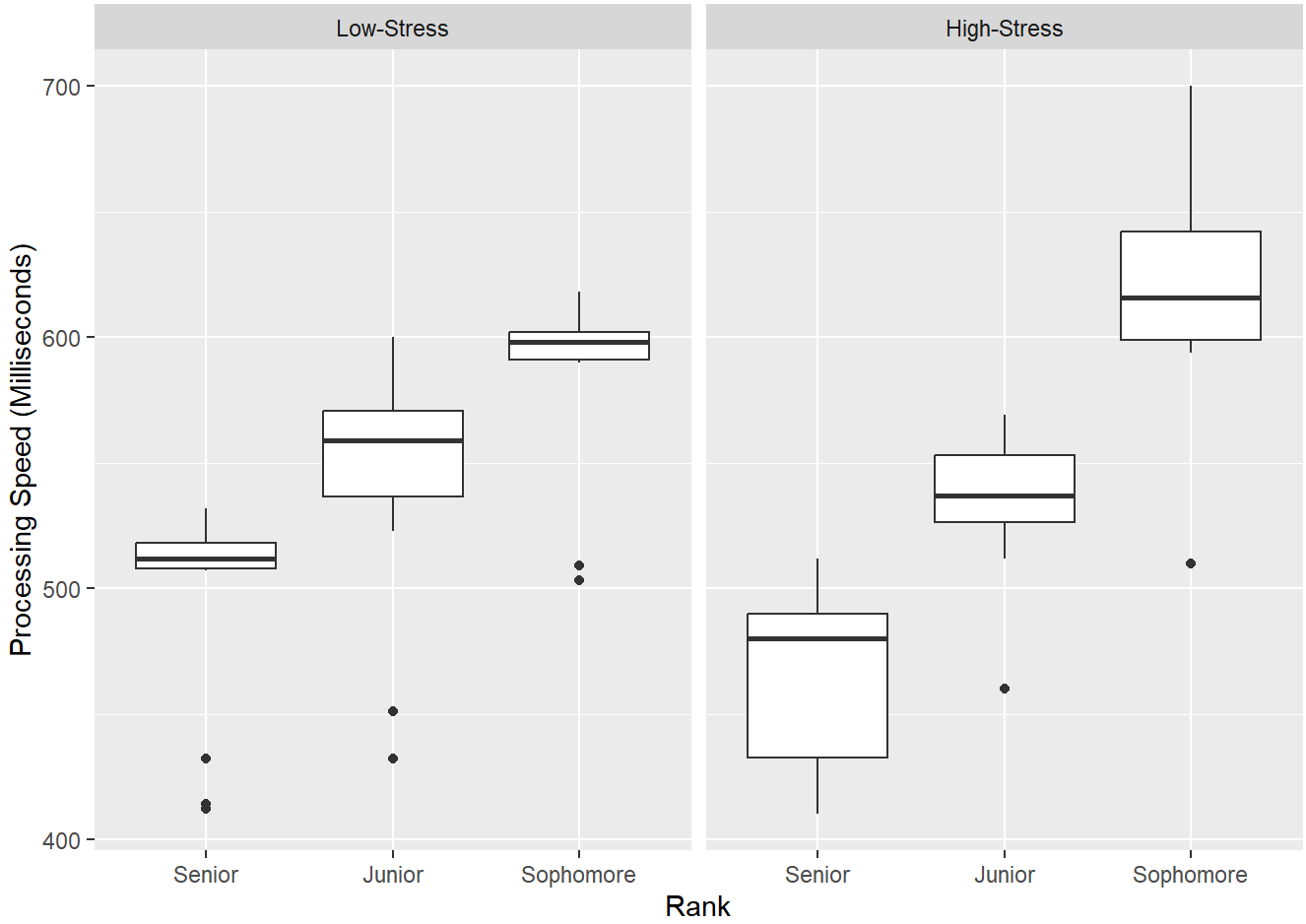

```
ggplot(data = dat_MF,
       mapping = aes(y = Time, x = Rank)) +
  geom_boxplot() +
    labs(y = "Processing Speed (Milliseconds)")
```



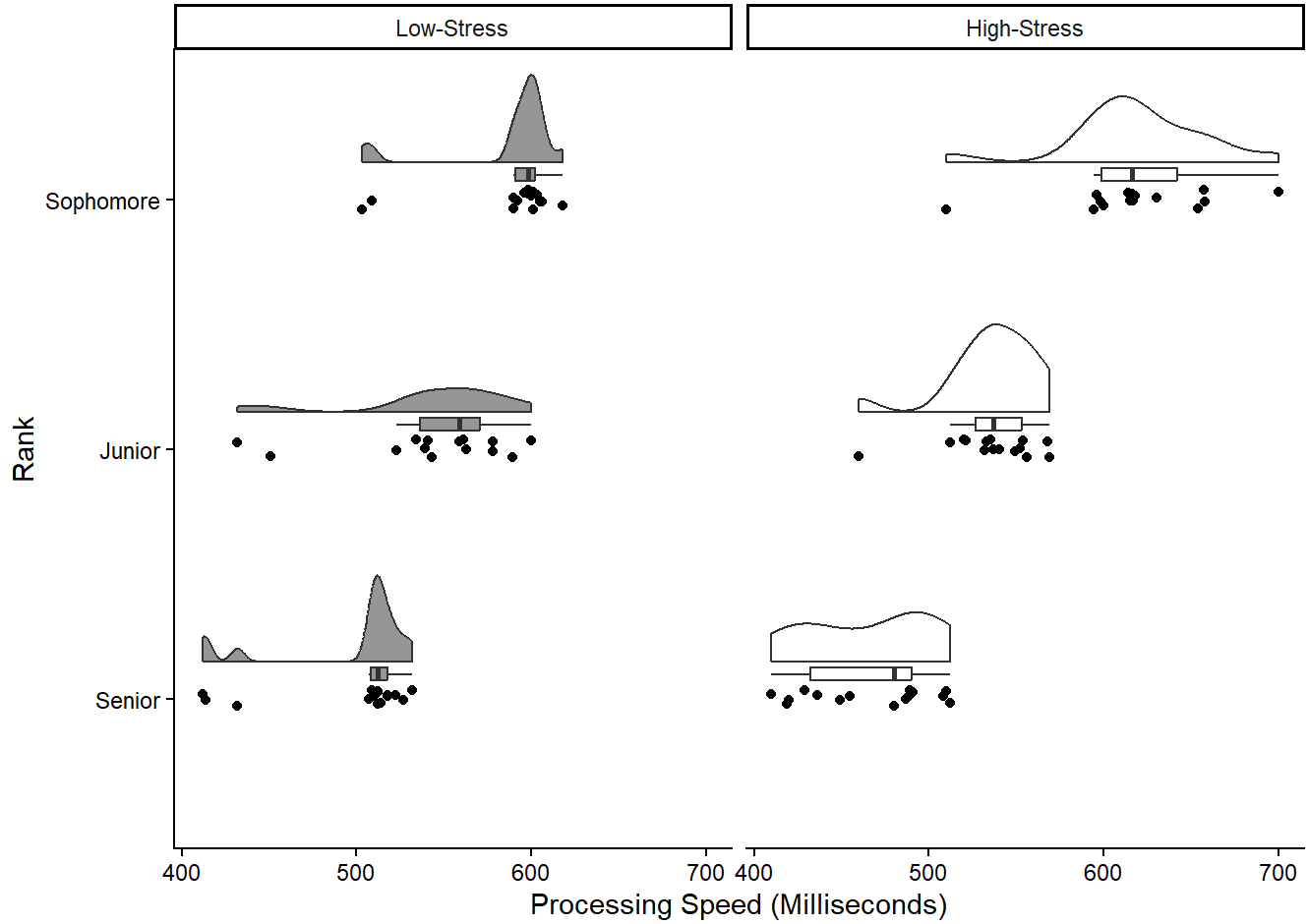

```
ggplot(data = dat_MF,
       mapping = aes(y = Time, x = Condition)) +
  geom_boxplot() +
  facet_wrap(~Rank)+
    labs(y = "Processing Speed (Milliseconds)")
```




```
ggplot(data = dat_MF,
       mapping = aes(y = Time, x = Rank)) +
  geom_boxplot() +
  facet_wrap(~Condition) +
    labs(y = "Processing Speed (Milliseconds)")
```




```
ggplot(data = dat_MF,
       aes(x = Rank, y = Time,
           fill = Condition))+
  geom_rain(show.legend=FALSE) +
  facet_wrap(~Condition)+
  scale_colour_grey(aesthetics = "fill",
                    start = .6,
                    end = 1) +
  coord_flip() +
  ylab("Processing Speed (Milliseconds)")+
  theme_classic()
```

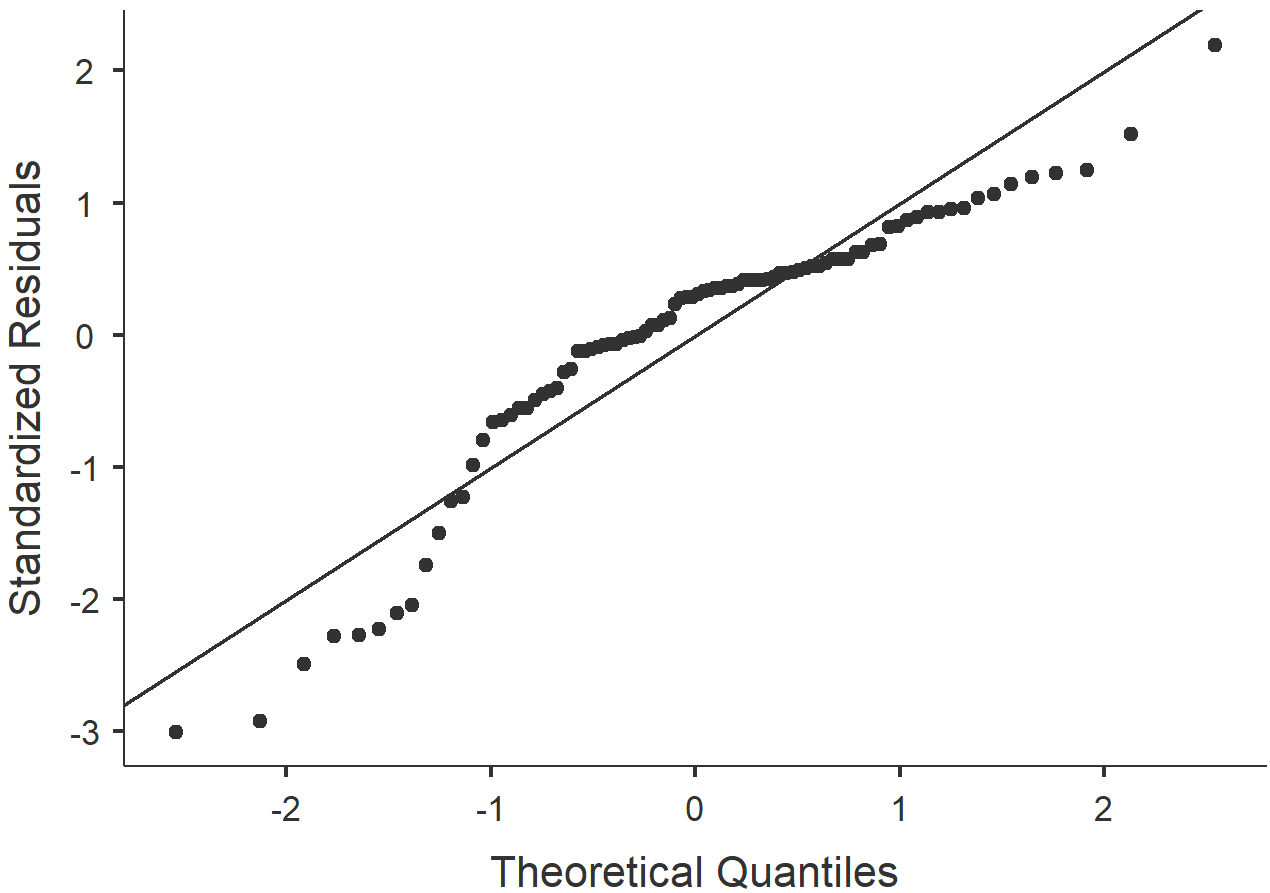


Mixed Factorial ANOVA | anovaRM() Omnibus Assumption Checks

```
anovaRM(data=dat_MFW,
        bs = list('Condition'),
        bsTerms = list('Condition'),
        rm = list(
             list(label = 'Rank',
                  levels = list('Sophomore',
                                'Junior',
                                'Senior'))),
        rmCells = list(
           list(measure = 'Sophomore',
                cell = 'Sophomore'),
           list(measure = 'Junior',
                cell = 'Junior'),
           list(measure = 'Senior',
                cell = 'Senior')),
        rmTerms= list('Rank'),
        leveneTest = TRUE,
        qq = TRUE,
        spherTests = TRUE)
```

```
##
##  REPEATED MEASURES ANOVA
##
##  Within Subjects Effects
##  ─────────────────────────────────────────────────────────────────────────────
##                      Sum of Squares    df    Mean Square    F            p
##  ─────────────────────────────────────────────────────────────────────────────
##    Rank                   222390.87     2    111195.4333    158.38122    < .0000001
##    Rank:Condition          14838.29     2      7419.1444     10.56746     0.0001277
##    Residual                39316.18    56       702.0746
##  ─────────────────────────────────────────────────────────────────────────────
##    Note. Type 3 Sums of Squares
##
##
##  Between Subjects Effects
##  ──────────────────────────────────────────────────────────────────────────────
##                 Sum of Squares    df    Mean Square    F             p
##  ──────────────────────────────────────────────────────────────────────────────
##    Condition          149.5111     1       149.5111    0.05004870    0.8246028
##    Residual         83644.7556    28      2987.3127
##  ──────────────────────────────────────────────────────────────────────────────
##    Note. Type 3 Sums of Squares
##
##
##  ASSUMPTIONS
##
##  Tests of Sphericity
##  ────────────────────────────────────────────────────────────────────────────
##            Mauchly's W    p            Greenhouse-Geisser ε    Huynh-Feldt ε
##  ────────────────────────────────────────────────────────────────────────────
##    Rank      0.6845379    0.0059961               0.7601891        0.7947804
##  ────────────────────────────────────────────────────────────────────────────
##
##
##  Homogeneity of Variances Test (Levene's)
##  ──────────────────────────────────────────────────────
##                 F             df1    df2    p
##  ──────────────────────────────────────────────────────
##    Sophomore    0.32438535      1     28    0.5735255
##    Junior       2.03565460      1     28    0.1647047
##    Senior       0.02578285      1     28    0.8735846
##  ──────────────────────────────────────────────────────
```

```
## Warning: The `size` argument of `element_line()` is deprecated as of ggplot2 3.4.0.
## ℹ Please use the `linewidth` argument instead.
## ℹ The deprecated feature was likely used in the jmvcore package.
##   Please report the issue at <https://github.com/jamovi/jmvcore/issues>.
## This warning is displayed once per session.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```



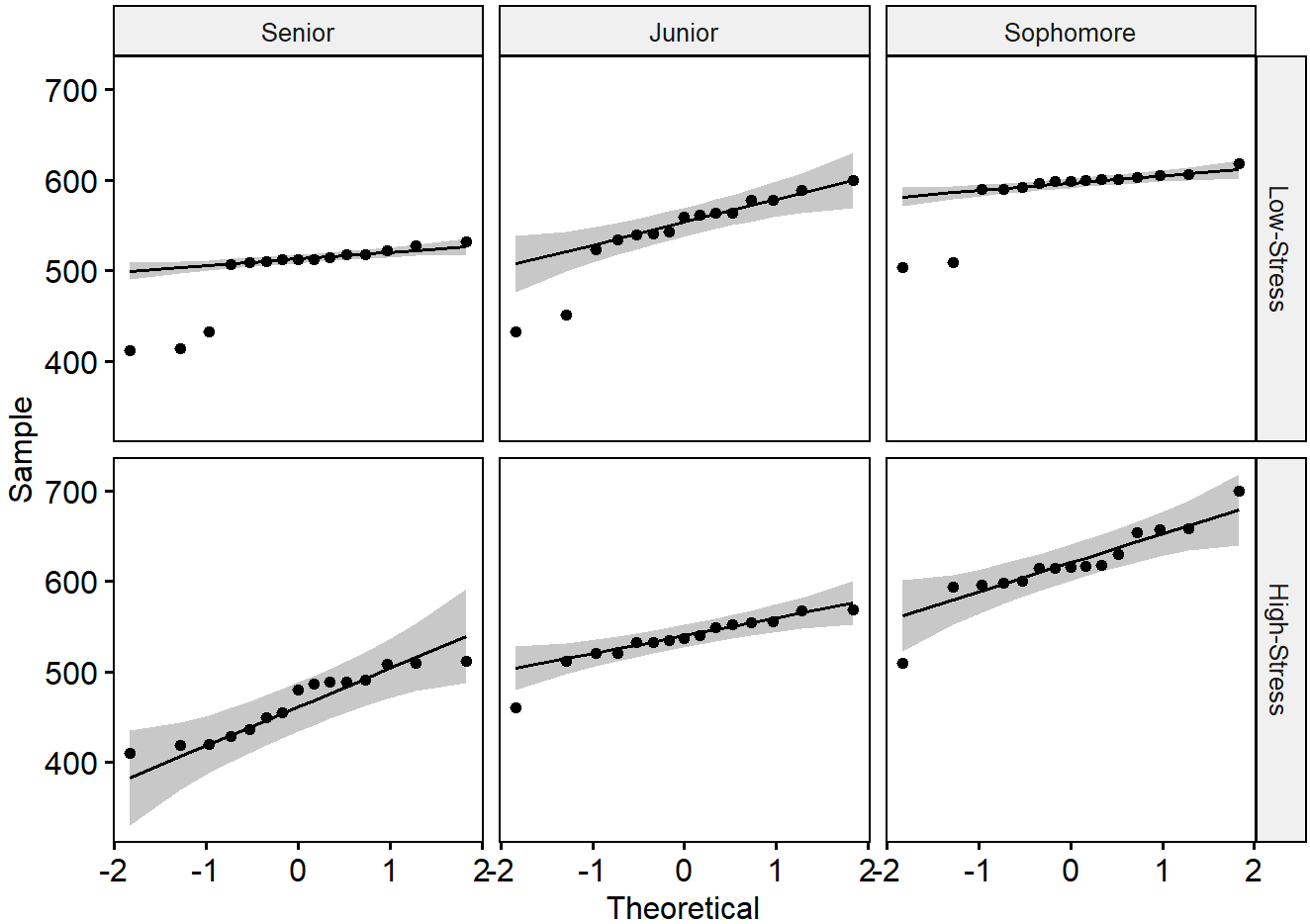
Mixed Factorial ANOVA | Group Level Assumption Checks

```
dat_MF %>%
  group_by(Condition, Rank) %>%
  shapiro_test(Time)
```

```
## # A tibble: 6 × 5
##   Condition   Rank      variable statistic         p
##   <fct>       <fct>     <chr>        <dbl>     <dbl>
## 1 Low-Stress  Senior    Time         0.672 0.000126
## 2 Low-Stress  Junior    Time         0.841 0.0131
## 3 Low-Stress  Sophomore Time         0.617 0.0000365
## 4 High-Stress Senior    Time         0.901 0.0977
## 5 High-Stress Junior    Time         0.875 0.0395
## 6 High-Stress Sophomore Time         0.899 0.0935
```

Mixed Factorial ANOVA | Group Level Assumption Checks

```
ggqqplot(dat_MF,
         'Time',
         facet.by = c("Condition","Rank"))
```



Mixed Factorial ANOVA | Omnibus Levene's Test Assumption Checks

```
dat_MFW %>%
  levene_test(averageTime ~ Condition,
              center = mean)
```

```
## # A tibble: 1 × 4
##     df1   df2 statistic     p
##   <int> <int>     <dbl> <dbl>
## 1     1    28      1.90 0.179
```

Mixed Factorial ANOVA | Group Level Assumption Checks

```
# Must be conducted ON A MATRIX!! Also, note that it is using the long version of the data!

box_m(as.data.frame(dat_MF$Time),
      dat_MF$Condition)
```

```
## # A tibble: 1 × 4
##   statistic p.value parameter method
##       <dbl>   <dbl>     <dbl> <chr>
## 1      3.31  0.0690         1 Box's M-test for Homogeneity of Covariance Matric…
```

Mixed Factorial ANOVA | Conducting the ANOVA

```
options(digits = 3)
anovaRM(data = dat_MFW,
        bs = list('Condition'),
        bsTerms = list('Condition'),
        rm = list(
             list(label = 'Rank',
                  levels = list('Sophomore',
                                'Junior',
                                'Senior'))),
        rmCells = list(
           list(measure = 'Sophomore',
                cell = 'Sophomore'),
           list(measure = 'Junior',
                cell = 'Junior'),
           list(measure = 'Senior',
                cell = 'Senior')),
        rmTerms = list('Rank'),
        spherCorr = c("none","HF"),
        postHoc = list('Rank', c('Condition','Rank')),
        postHocCorr = 'bonf',
        emMeans = list('Rank','Condition', c('Rank','Condition')),
        emmPlots = FALSE,
        emmTables = TRUE,
        effectSize = 'partEta')
```

```
##
##  REPEATED MEASURES ANOVA
##
##  Within Subjects Effects
##  ──────────────
```

```
##    Condition      Rank              Condition      Rank         Mean Difference    SE       d
f      t         p-bonferroni
## ─────────────────────────────────────────────────────────────────────────────────────────────
──────────────────────────────
##    Low-Stress     Sophomore    -    Low-Stress     Junior                43.73     8.46    2
8.0     5.168    < .001
##                                 -    Low-Stress     Senior                90.60    12.08    2
8.0     7.499    < .001
##                                 -    High-Stress    Sophomore            -31.13    13.93    2
8.0    -2.236          0.503
##                                 -    High-Stress    Junior                51.47    13.92    2
8.0     3.698          0.014
##                                 -    High-Stress    Senior               121.73    14.00    2
8.0     8.694    < .001
##                    Junior       -    Low-Stress     Senior                46.87     7.95    2
8.0     5.893    < .001
##                                 -    High-Stress    Sophomore            -74.87    13.92    2
8.0    -5.380    < .001
##                                 -    High-Stress    Junior                 7.73    13.91    2
8.0     0.556          1.000
##                                 -    High-Stress    Senior                78.00    13.99    2
8.0     5.574    < .001
##                    Senior       -    High-Stress    Sophomore           -121.73    14.00    2
8.0    -8.694    < .001
##                                 -    High-Stress    Junior               -39.13    13.99    2
8.0    -2.797          0.138
##                                 -    High-Stress    Senior                31.13    14.08    2
8.0     2.211          0.530
##    High-Stress    Sophomore    -    High-Stress    Junior                82.60     8.46    2
8.0     9.761    < .001
##                                 -    High-Stress    Senior               152.87    12.08    2
8.0    12.653    < .001
##                    Junior       -    High-Stress    Senior                70.27     7.95    2
8.0     8.835    < .001
## ─────────────────────────────────────────────────────────────────────────────────────────────
──────────────────────────────
##
##
##  ESTIMATED MARGINAL MEANS
##
##  RANK
##
##  Estimated Marginal Means - Rank
##  ──────────────────────────────────────────────
##    Rank          Mean    SE      Lower    Upper
##  ──────────────────────────────────────────────
##    Sophomore      603    6.96      589      617
##    Junior         540    6.95      525      554
##    Senior         481    7.04      467      496
##  ──────────────────────────────────────────────
##
##
```

```
##  CONDITION
##
##  Estimated Marginal Means - Condition
##  ──────────────────────────────────────────────────
##    Condition      Mean    SE      Lower    Upper
##  ──────────────────────────────────────────────────
##    Low-Stress      543    8.15      526      559
##    High-Stress     540    8.15      523      557
##  ──────────────────────────────────────────────────
##
##
##  RANK:CONDITION
##
##  Estimated Marginal Means - Rank:Condition
##  ──────────────────────────────────────────────────────────────
##    Condition      Rank         Mean    SE      Lower    Upper
##  ──────────────────────────────────────────────────────────────
##    Low-Stress     Sophomore     587    9.85      567      608
##                   Junior        544    9.83      523      564
##                   Senior        497    9.95      476      517
##    High-Stress    Sophomore     618    9.85      598      639
##                   Junior        536    9.83      516      556
##                   Senior        466    9.95      445      486
##  ──────────────────────────────────────────────────────────────
```

Mixed Factorial ANOVA | ANOVA Output: Interaction Simple Effects

```
one.way <- dat_MF %>%
  group_by(Condition) %>%
  anova_test(dv = Time,
             wid = Subject,
             within = Rank,
             effect.size = 'ges') %>%
  get_anova_table(correction = "none") %>%
  adjust_pvalue(method = "bonferroni")

one.way
```

```
## # A tibble: 2 × 9
##   Condition   Effect   DFn   DFd     F        p `p<.05`   ges    p.adj
##   <fct>       <chr>  <dbl> <dbl> <dbl>    <dbl> <chr>   <dbl>    <dbl>
## 1 Low-Stress  Rank       2    28  86.5 1.04e-12 *       0.469 2.08e-12
## 2 High-Stress Rank       2    28  83.8 1.52e-12 *       0.768 3.04e-12
```

```
one.way2 <- dat_MF %>%
  group_by(Rank) %>%
  anova_test(dv = Time,
             between = Condition,
             effect.size = 'ges') %>%
  get_anova_table() %>%
  adjust_pvalue(method = "bonferroni")

one.way2
```

```
## # A tibble: 3 × 9
##   Rank      Effect      DFn   DFd     F     p `p<.05`   ges p.adj
##   <fct>     <chr>     <dbl> <dbl> <dbl> <dbl> <chr>   <dbl> <dbl>
## 1 Senior    Condition     1    28 4.89  0.035 "*"     0.149 0.105
## 2 Junior    Condition     1    28 0.309 0.583 ""      0.011 1
## 3 Sophomore Condition     1    28 5.00  0.034 "*"     0.151 0.102
```

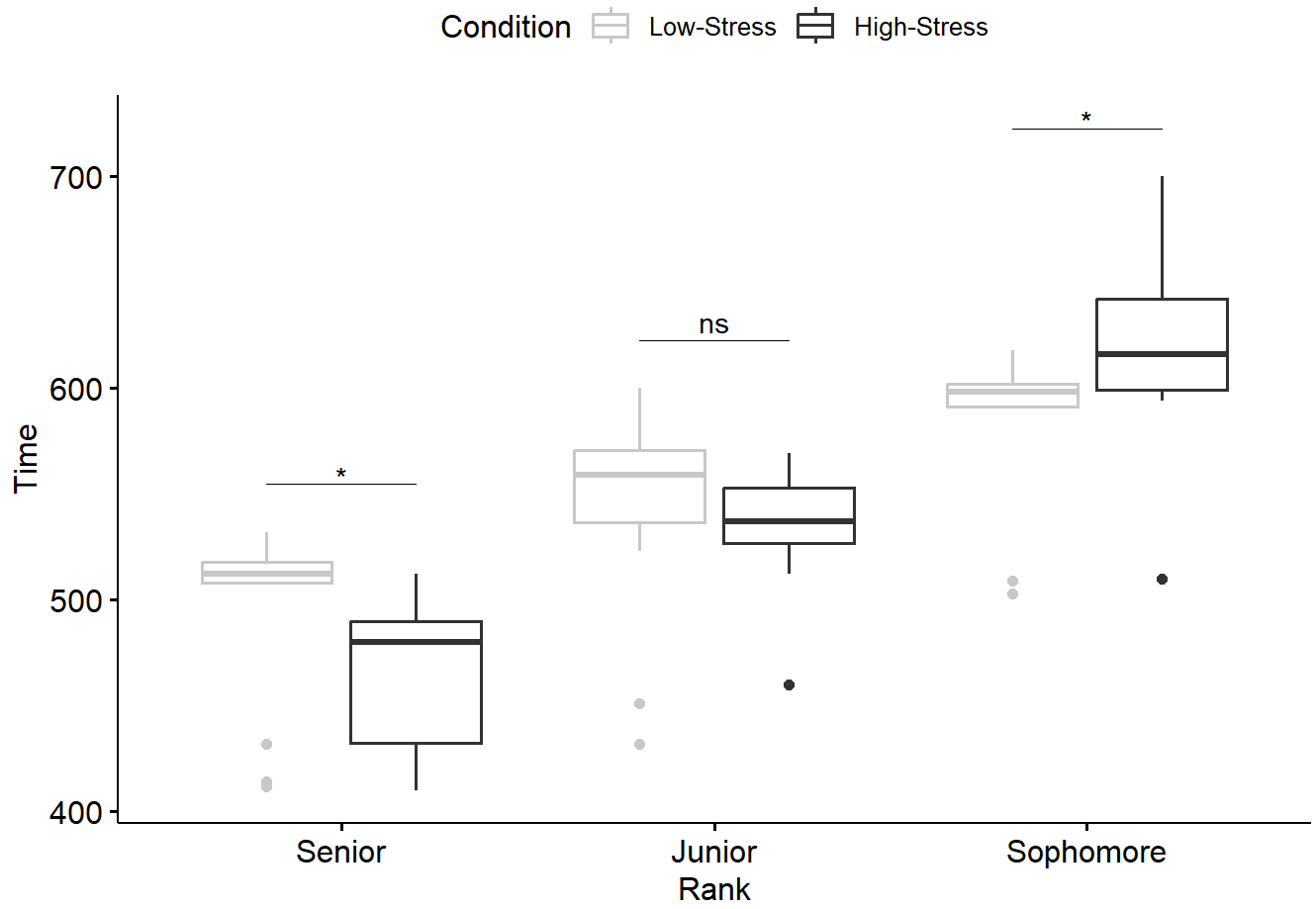
Demonstrates that there are significant differences between the Ranks for participants in the Low-Stress [$F(2,28) = 86.50$, $p < .001$, $\eta^2 = .47$] and High-Stress [$F(2,28) = 83.79$, $p < .001$, $\eta^2 = .77$] conditions—furthermore, there is an issue of Sphericity demonstrated for the High-Stress condition (note the degrees of freedom!). However, this requires that we conduct additional post hoc tests.

On the other hand, based on Bonferroni corrected p-values there were no statistically significant differences between stress conditions for students regardless of school rank: sophomore year [$F(1,28) = 5.00$, $p = .10$, $\eta_p^2 = .15$], junior year [$F(1,28) = 0.31$, $p = .58$, $\eta_p^2 = .01$], and senior year [$F(1,28) = 4.89$, $p = .11$, $\eta_p^2 = .15$]. The lack of statistical significance here is probably due to the sample size being under-powered, look at the partial eta-square values.

Mixed Factorial ANOVA | Professional ANOVA Visualization Code

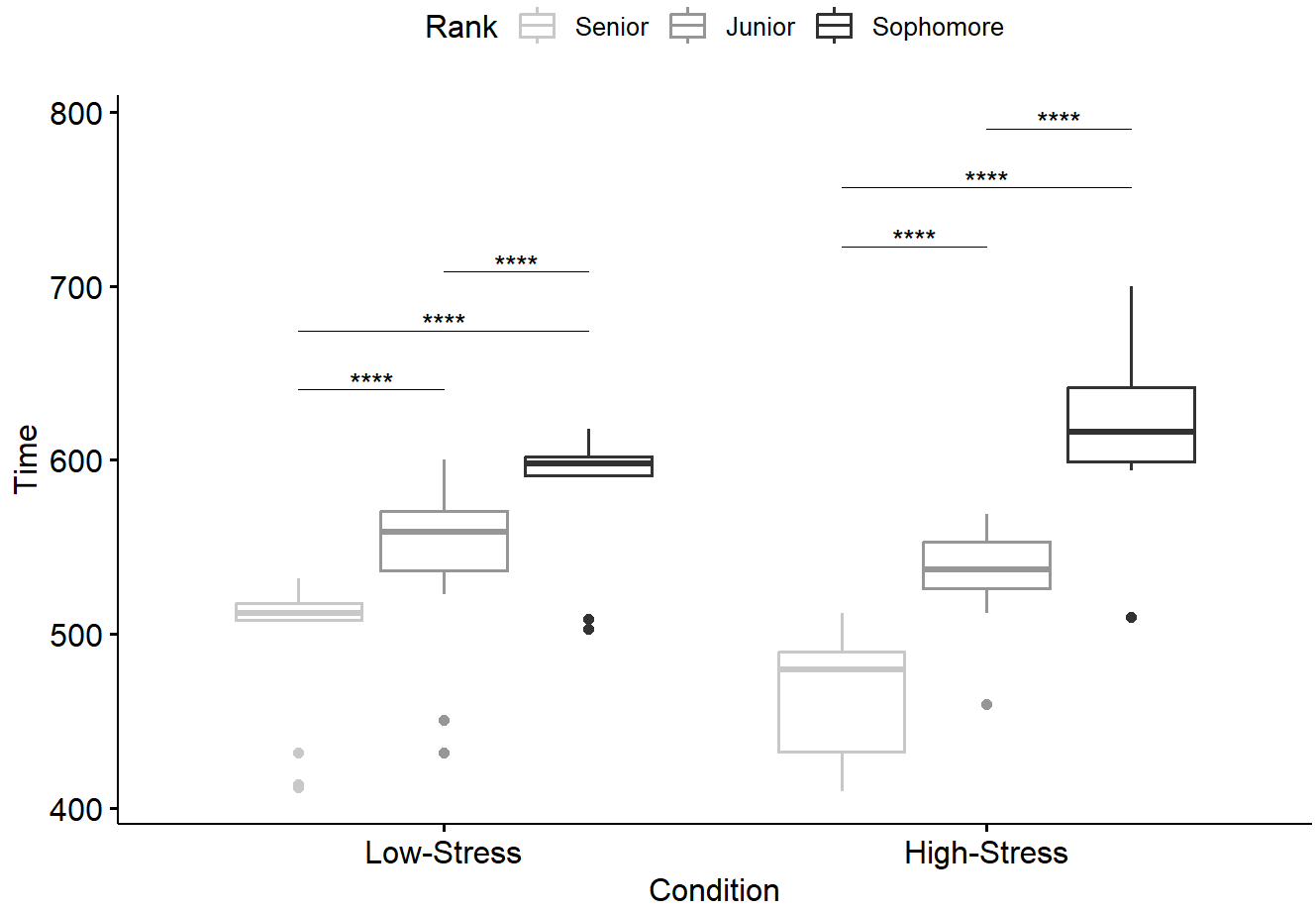
```
pwc1 <- dat_MF %>%
  group_by(Rank) %>%
  pairwise_t_test(Time~Condition,
                  p.adjust.method = "bonferroni") %>%
  add_xy_position(x = "Rank",
                  group = "Condition")

ggboxplot(dat_MF,
          x = "Rank",
          y = "Time",
          color = "Condition",
          palette = "grey")+
  stat_pvalue_manual(pwc1,
                     hide.ns = FALSE,
                     tip.length = 0)
```




```
pwc2 <- dat_MF %>%
  group_by(Condition) %>%
  pairwise_t_test(Time~Rank,
                  p.adjust.method = "bonferroni",
                  paired = TRUE) %>%
  add_xy_position(x = "Condition",
                  group = "Rank")

ggboxplot(dat_MF,
          x = "Condition",
          y = "Time",
          color = "Rank",
          palette = "grey")+
  stat_pvalue_manual(pwc2,
                     hide.ns = FALSE,
                     tip.length = 0)
```



Mixed Factorial ANOVA | APA Style ANOVA Tables

```
res.aov <- dat_MF %>%
  ezANOVA(dv = .(Time),
          wid = .(Subject),
          between= .(Condition),
          within = .(Rank),
          detailed = TRUE)

res.aov$ANOVA$ges <- res.aov$ANOVA$SSn/(res.aov$ANOVA$SSn+res.aov$ANOVA$SSd)

res.aov$ANOVA<-res.aov$ANOVA[c(1:3,6:7,9)]


apa.ezANOVA.table(res.aov,
                  table.title = "Mixed Factorial ANOVA",
                  correction="HF",
                  table.number=7)
```

```
##
##
## Table 7
##
## Mixed Factorial ANOVA
##
##         Predictor df_num df_den Epsilon       F    p  ges
##       (Intercept)   1.00  28.00         8826.42 .000 1.00
##         Condition   1.00  28.00            0.05 .825  .00
##              Rank   1.59  44.51    0.79  158.38 .000  .85
##  Condition x Rank   1.59  44.51    0.79   10.57 .000  .27
##
## Note. df_num indicates degrees of freedom numerator. df_den indicates degrees of freedom deno
minator.
## Epsilon indicates Huynh-Feldt multiplier for degrees of freedom,
## p-values and degrees of freedom in the table incorporate this correction.
## ges indicates generalized eta-squared.
##
```

```
#Generate Marginal Means Table
apa.2way.table(Rank,
               Condition,
               Time,
               dat_MF,
               table.number=8,
               show.conf.interval = TRUE)
```

```
##
##
## Table 8
##
## Means and standard deviations for Time as a function of a 3(Rank) X 2(Condition) design
##
##                              M         M_95%_CI    SD
##   Condition:Low-Stress
##                    Rank
##                  Senior 496.73 [474.12, 519.35] 40.84
##                  Junior 543.60 [517.76, 569.44] 46.66
##               Sophomore 587.33 [568.63, 606.04] 33.77
##
##  Condition:High-Stress
##                    Rank
##                  Senior 465.60 [445.59, 485.61] 36.13
##                  Junior 535.87 [520.96, 550.77] 26.91
##               Sophomore 618.47 [595.18, 641.76] 42.05
##
## Note. M and SD represent mean and standard deviation, respectively.
## LL and UL indicate the lower and upper limits of the
## 95% confidence interval for the mean, respectively.
## The confidence interval is a plausible range of population means
## that could have created a sample mean (Cumming, 2014).
```